

D878

General Ecology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

In a grassland covering 40 hectares, where rain fell on 12 of the last 30 days, producers capture 50,000 kcal of energy. About how much reaches the secondary consumers?

- A. 50 kcal
- B. 500 kcal
- C. 5,000 kcal
- D. 40,500 kcal

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Tertiary consumers in a food chain receive about 20 kcal. Using the 10% rule, about how much energy did the producers capture?

- A. 20,000 kcal
- B. 200,000 kcal
- C. 2,000 kcal
- D. 0.02 kcal

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A team surveys a pond near a trailhead with a new wooden boardwalk, recording algae, water temperature, dissolved oxygen, and fallen leaves. The boardwalk was built with volunteer labor. Which factor on the list is abiotic?

- A. Algae
- B. Fallen leaves
- C. Dissolved oxygen
- D. The wooden boardwalk

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A farmer rotates soybeans into a field that has grown corn for four years, and the county extension office reports soil nitrogen rising afterward. The field is 80 acres. What process explains the gain?

- A. Soybean leaves absorb N_2 directly from the air.
- B. Bacteria in soybean root nodules fix N_2 from the air into ammonia.
- C. Decomposing corn stalks create new nitrogen.
- D. Denitrifying bacteria add nitrogen to the soil.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Which process returns nitrogen from the soil to the atmosphere as N_2 ?

- A. Nitrification
- B. Ammonification
- C. Nitrogen fixation
- D. Denitrification

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A deer population in a state forest with a new visitor app starts the year at 2,000. During the year there are 300 births, 180 deaths, 40 deer moving in, and 60 moving out. The app logged 12,000 visits. What is the population at year's end?

- A. 2,580
- B. 2,140
- C. 2,100
- D. 2,160

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A population of 500 grows by 25 in one year. What is its per-capita growth rate?

- A. 25 per year
- B. 20 per year
- C. 0.05 per year
- D. 0.5 per year

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A marine species with a shimmering blue shell releases millions of eggs into open water each season and gives no parental care. Its range covers about 2,000 km of coastline. Which survivorship curve fits?

- A. Type II
- B. Type III
- C. Type I, since millions of eggs means most survive
- D. Type I

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Twenty years of data from an invented northern research station, recorded by rotating graduate teams, show that vole numbers peaked in years 4, 8, and 12. Owls feed mostly on voles. When did owl numbers most likely peak?

- A. About a year after each vole peak
- B. At the lowest vole years
- C. A year before each vole peak
- D. Never; predator numbers always stay constant

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Meadow A has 5 plant species, each making up 20% of the plants. Meadow B also has 5 species, but one makes up 92% of the plants. Meadow B has twice as many total plants and is closer to the road. Which meadow has higher species diversity?

- A. Meadow B, because one species is thriving
- B. They are equal, because each has 5 species
- C. Meadow B, because it has more total plants
- D. Meadow A, because its species are evenly represented

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A persistent, fat-soluble pesticide washes into a lake used by a popular fishing club with 400 members. The food chain is plankton → small fish → large fish → fish-eating birds. Which organisms will carry the highest concentration of the pesticide in their tissues?

- A. The fish-eating birds
- B. The large fish the club members catch
- C. The plankton, where the pesticide first enters
- D. All levels equally, since the toxin spreads evenly

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A toxin's concentration increases about tenfold at each trophic level. Plankton contain 0.04 ppm. What is the expected concentration in birds that eat large fish, which eat small fish, which eat plankton?

- A. 0.00004 ppm
- B. 4 ppm
- C. 0.04 ppm
- D. 40 ppm

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

In a grassland ~~covering 40 hectares, where rain fell on 12 of the last 30 days,~~ producers capture 50,000 kcal of energy. About how much reaches the secondary consumers?

Target word: energy at the secondary consumer level **Decides:** Two transfers: producers → primary → secondary.

A	Confusion set	Applies three transfers; that is the tertiary level.
B	Correct	$50,000 \times 0.10 \times 0.10 = 500$ kcal.
C	Dropped element	Applies only one transfer; that is the primary-consumer level.
D	Inverted operation	Subtracts 10% twice instead of keeping 10%.

2. Answer A Concept 1 • transfer

Tertiary consumers in a food chain receive about 20 kcal. Using the 10% rule, about how much energy did the producers capture?

Target word: solve backward for producer energy **Decides:** Three transfers separate producers and tertiary consumers.

A	Correct	$20 \times 10 \times 10 \times 10 = 20,000$ kcal.
B	Confusion set	Counts four transfers.
C	Dropped element	Counts only two transfers.
D	Inverted operation	Divides by 10 at each step instead of multiplying.

3. Answer C Concept 2

A team surveys a pond ~~near a trailhead with a new wooden boardwalk,~~ recording algae, water temperature, dissolved oxygen, and fallen leaves. ~~The boardwalk was built with volunteer labor.~~ Which factor on the list is abiotic?

Target word: identify the abiotic factor **Decides:** Dissolved oxygen is a nonliving chemical condition.

A	Inverted operation	Living producers are biotic.
B	Confusion set	Once-living material counts as biotic.
C	Correct	A chemical property of the water, not an organism.
D	Camouflage	A human structure, not one of the pond factors being sorted.

4. Answer B Concept 3

A farmer rotates soybeans into a field ~~that has grown corn for four years,~~ and the county extension office reports soil nitrogen rising afterward. ~~The field is 80 acres.~~ What process explains the gain?

Target word: source of new usable nitrogen **Decides:** Soybean root nodules host nitrogen-fixing bacteria.

A	Confusion set	Plants cannot use N_2 directly.
B	Correct	Nitrogen fixation adds usable nitrogen to soil.
C	Adjacent category	Decomposition recycles existing nitrogen; it does not add new nitrogen from the air.
D	Inverted operation	Denitrification removes nitrogen, returning N_2 to the air.

5. Answer D Concept 3 · transfer

Which process returns nitrogen from the soil to the atmosphere as N_2 ?

Target word: name the reverse step **Decides:** N_2 back to the air = denitrification.

A	Confusion set	Converts ammonia to nitrite and nitrate within the soil.
B	Adjacent category	Releases ammonia from dead matter in the soil.
C	Inverted operation	Moves N_2 from the air into the soil.
D	Correct	Denitrifying bacteria convert nitrate to N_2 gas.

6. Answer C Concept 4

A deer population in a state forest ~~with a new visitor app~~ starts the year at 2,000. During the year there are 300 births, 180 deaths, 40 deer moving in, and 60 moving out. ~~The app logged 12,000 visits.~~ What is the population at year's end?

Target word: end-of-year population **Decides:** $\text{Change} = 300 + 40 - 180 - 60 = +100$.

A	Confusion set	Adds all four numbers as gains.
B	Inverted operation	Treats immigration as a loss and emigration as a gain.
C	Correct	$2,000 + 100 = 2,100$.
D	Dropped element	Leaves out emigration.

7. Answer C Concept 4 · transfer

A population of 500 grows by 25 in one year. What is its per-capita growth rate?

Target word: solve for per-capita rate **Decides:** $r = \text{change} / \text{starting } N$.

A	Wrong stem number	The total change, not the per-capita rate.
B	Inverted operation	$500 / 25$ flips the ratio.
C	Correct	$25 / 500 = 0.05$.
D	Dropped element	Drops a decimal place; that would be 250 new individuals.

8. Answer B Concept 5

A marine species ~~with a shimmering blue shell~~ releases millions of eggs into open water each season and gives no parental care. ~~Its range covers about 2,000 km of coastline.~~ Which survivorship curve fits?

Target word: survivorship type **Decides:** Many offspring and no care means most die young.

A	Adjacent category	Type II has a steady death rate across ages.
B	Correct	Very high early mortality, few survive to adulthood.
C	Confusion set	Many offspring usually signals high early loss, not high survival.
D	Inverted operation	Type I fits few offspring with much care.

9. Answer A Concept 6

Twenty years of data from an invented northern research station, recorded by rotating graduate teams, show that vole numbers peaked in years 4, 8, and 12. Owls feed mostly on voles. When did owl numbers most likely peak?

Target word: timing of predator peaks **Decides:** Predators respond to prey with a lag.

A	Correct	More prey supports more predator offspring, which follow later.
B	Confusion set	Owl lows follow vole lows; they do not peak then.
C	Inverted operation	Predators cannot rise before their food supply does.
D	Absolute language	Predator numbers track prey availability.

10. Answer D Concept 7

Meadow A has 5 plant species, each making up 20% of the plants. Meadow B also has 5 species, but one makes up 92% of the plants. ~~Meadow B has twice as many total plants and is closer to the road.~~ Which meadow has higher species diversity?

Target word: compare diversity **Decides:** Same richness; A is far more even.

A	Inverted operation	Dominance by one species lowers evenness and diversity.
B	Dropped element	Ignores evenness.
C	Camouflage	Total abundance does not measure diversity.
D	Correct	Equal richness plus higher evenness means higher diversity.

11. Answer A Concept 8

A persistent, fat-soluble pesticide washes into a lake ~~used by a popular fishing club with 400 members.~~ The food chain is plankton → small fish → large fish → fish-eating birds. Which organisms will carry the highest concentration of the pesticide in their tissues?

Target word: where the toxin concentrates **Decides:** Biomagnification peaks at the top predator.

A	Correct	They eat many contaminated fish, and the toxin builds up at each level.
B	Camouflage	High, but below the birds that eat them.
C	Inverted operation	Concentration rises, not falls, up the chain.
D	Absolute language	Persistent toxins build up rather than spread evenly.

12. Answer D Concept 8 · transfer

A toxin's concentration increases about tenfold at each trophic level. Plankton contain 0.04 ppm. What is the expected concentration in birds that eat large fish, which eat small fish, which eat plankton?

Target word: concentration at the top **Decides:** Three steps up: $\times 10$ three times.

A	Inverted operation	Divides by 10 at each step.
B	Dropped element	Counts only two steps.
C	Absolute language	Assumes concentration stays the same at every level.
D	Correct	$0.04 \times 10 \times 10 \times 10 = 40$ ppm.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	3	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	12	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	The 10% rule for energy	About 10% of the energy at one trophic level passes to the next; the rest is lost mostly as heat.	Subtracting 10% instead of keeping 10%, or miscounting the number of transfers.	Producer energy and a question about a higher level.
2	Biotic vs. abiotic factors	Biotic factors are living or once-living parts of an ecosystem; abiotic factors are nonliving physical and chemical conditions.	Classifying things that come from living organisms, or that affect them, as abiotic (or the reverse).	A list of ecosystem factors to sort.
3	Key nitrogen-cycle steps	Nitrogen-fixing bacteria turn N_2 into ammonia; nitrifying bacteria make nitrate; denitrifying bacteria return N_2 to the air. Most organisms cannot use N_2 directly.	Thinking plants absorb N_2 from the air, or swapping fixation and denitrification.	Nitrogen moving between air, soil, and organisms.
4	Population change accounting	Change in N = births + immigration – deaths – emigration; per-capita rate = change / starting N .	Dropping migration terms or flipping their signs.	Counts of births, deaths, and movement in or out.
5	Survivorship curves	Type I: most survive to old age (few offspring, much care); Type II: steady death rate; Type III: most die young (many offspring, little care).	Assuming many offspring means high survival.	Offspring number and parental care described.
6	Predator-prey cycles	Predator numbers rise after prey numbers rise and fall after prey fall, so predator peaks lag prey peaks.	Placing the predator peak before or at the same time as the prey peak.	Linked population graphs over years.
7	Species richness vs. evenness	Diversity reflects both the number of species (richness) and how evenly individuals are spread among them (evenness).	Judging diversity by species count or total individuals alone.	Two communities with the same species count but different proportions.
8	Biomagnification	Persistent, fat-soluble toxins become more concentrated at each higher trophic level, so top predators carry the highest levels.	Thinking toxin concentration is highest where it enters, or evens out.	A pesticide or heavy metal moving through a food chain.

Course scope (public description): General Ecology offers an in-depth exploration of how ecosystems function and interact. Candidates will study both biotic and abiotic factors that play crucial roles in sustaining ecological balance. Key topics include the flow of energy through food chains and webs, nutrient cycling, and the dynamics of populations—covering growth models, carrying capacity, and factors influencing population size and structure. The course examines species interactions such as competition, predation, and symbiosis, and explores community structure and biodiversity. Candidates will learn about primary and secondary succession and how these processes shape ecosystems over time. Additionally, the course addresses conservation principles and practices, with a focus on analyzing human impacts on ecosystems and exploring strategies for sustainability. Through this course, candidates will gain a comprehensive understanding of the intricate web of life that sustains our planet.